Statistics & Applications Project – Fall 1405

Phase 1: Study Design, Data, and Descriptive Statistics

Track A – NHANES 2021–2023

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Working research question: Is mean systolic blood pressure among U.S. adults (20+) associated with usual sleep duration, sedentary time, smoking status, and BMI, after accounting for age and sex?

This report, the companion notebook (notebooks/phase1_exploration.ipynb), and the reusable pipeline (src/analysis.py) were produced together so that all numbers below are reproducible end-to-end from the raw CDC files listed in data/README.md.

1. Data Source and Collection Mechanism

Official source: the National Health and Nutrition Examination Survey (NHANES), conducted by the U.S. National Center for Health Statistics (NCHS), part of the CDC. This project uses the August 2021–August 2023 cycle (file suffix *_L*), a combined, non-standard cycle released to bridge the data collection gap caused by the COVID-19 pandemic (data collection was suspended for part of 2020). Because of this, the cycle's sample weights (WTMEC2YR, WTINT2YR) are already scaled for this single two-year release and require no further adjustment.

Collection mechanism: a two-stage design — (1) an in-home interview covering demographics and lifestyle questionnaires (sleep, physical activity, smoking), and (2) a physical examination at a Mobile Examination Center (MEC), where trained staff measure body size and blood pressure with an oscillometric device. All files used are public-use and required no special access application.

File	Component
DEMO_L	Demographics & sample weights (base file)
BPXO_L	Blood pressure - oscillometric measurements
BMX_L	Body measures (height, weight, BMI, waist)
SLQ_L	Sleep disorders questionnaire
PAQ_L	Physical activity questionnaire
SMQ_L	Smoking - cigarette use questionnaire

2. Observation Unit and Data Structure

Each row represents one survey respondent, uniquely identified by SEQN across every component file. The data are stored wide (one row per person); the six component files were merged on SEQN. There is no repeated-visit hierarchy in this subset; the only within-row repetition is three consecutive blood-pressure readings (BPXOSY1–3), averaged into the main response variable.

3. Sampling, Selection, and Generalizability

NHANES uses a complex multistage, stratified, clustered probability design with deliberate oversampling of specific subgroups (e.g., older adults, certain race/ethnicity groups, lower-income persons) to improve precision for those subgroups. This is an observational cross-sectional study (random sampling from a population, not random assignment to treatment). Consequently, respondents do not represent the population equally; the MEC sample weight corrects for this (Section A2/A3). Weighted results generalize to the U.S. civilian noninstitutionalized population aged 20+ during 2021–2023, not to institutionalized populations or to other countries.

4. Classification of Key Variables

Variable	Description	Type	Scale
sbp_mean	Mean systolic BP (3 readings)	Quantitative, continuous	Ratio
bmi	Body mass index	Quantitative, continuous	Ratio
sleep_hours	Usual weekday sleep duration	Quantitative, continuous	Ratio

Variable	Description	Type	Scale
sedentary_min	Daily sedentary minutes	Quantitative, continuous	Ratio
age_years	Age in years	Quantitative, discrete	Ratio
age_group	Age group (20-39/40-59/60+)	Qualitative	Ordinal
sex	Sex	Qualitative	Nominal (binary)
education	Education level	Qualitative	Ordinal
smoking_status	Smoking status (4 levels)	Qualitative	Nominal

5. Data Quality Audit

NHANES refused/don't-know sentinel codes (e.g., 7, 9, 7777, 9999 depending on the variable) in PAD680, SMQ040, SMQ020 and DMDEDUC2 were recoded to missing before analysis, since they are not valid numeric responses and would otherwise severely distort means (e.g., a sentinel of 7777 minutes of sedentary time). No rows were dropped at this stage — missing values are simply ignored locally in each computation.

variable	n_missing	pct_missing	n_valid
sbp_mean	0	0.0	5824
bmi	58	1.0	5766
sleep_hours	59	1.01	5765
sedentary_min	42	0.72	5782
age_years	0	0.0	5824
sex	0	0.0	5824
education	3	0.05	5821
smoking_status	7	0.12	5817

Plausibility check: no systolic BP values outside 70–250 mmHg, no BMI outside 10–80, and no sleep duration outside 2–16 hours were found in the analytic sample, so no observations were removed as implausible outliers in Phase 1.

6. Frequency Tables (Categorical Variables)

sex

sex	freq	pct
Female	3178	54.6
Male	2646	45.4

age_group

age_group	freq	pct
60+	2691	46.2
40-59	1681	28.9
20-39	1452	24.9

smoking_status

smoking_status	freq	pct
Never smoker	3396	58.3
Former smoker	1544	26.5
Current, every day	709	12.2

smoking_status	freq	pct
Current, some days	168	2.9
Missing	7	0.1

7. Center and Spread of Key Quantitative Variables

variable	n	mean	sd	median	iqr	min	max
sbp_mean	5824	123.03	18.08	120.67	22.0	70.0	232.33
bmi	5766	29.73	7.15	28.4	8.8	11.1	68.9
sleep_hours	5765	7.7	1.58	8.0	1.5	2.0	14.0
sedentary_min	5782	365.58	210.38	300.0	300.0	0.0	1380.0
age_years	5824	54.09	17.04	58.0	28.0	20.0	80.0

8. Visualizations

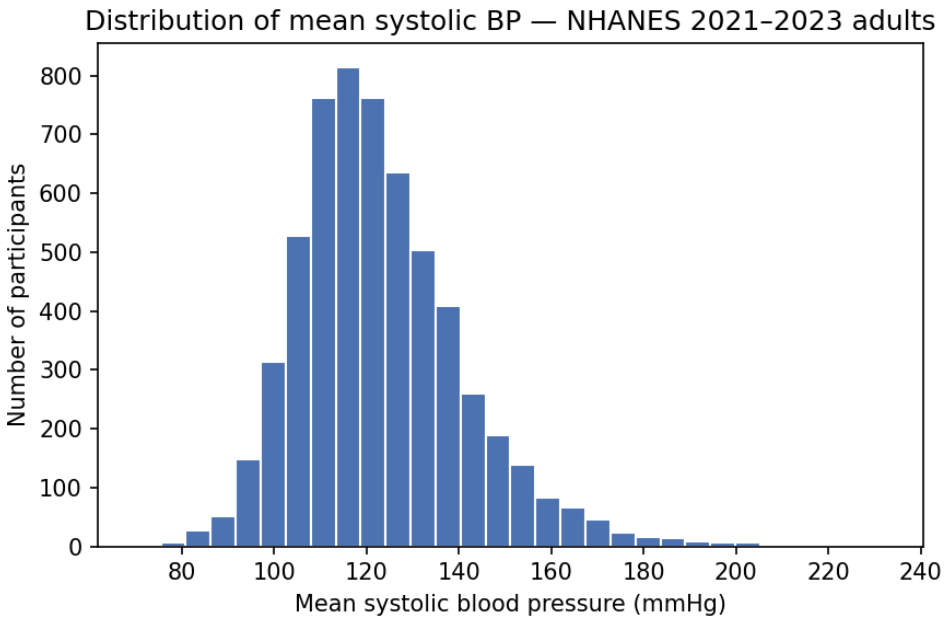


Figure 1. Distribution of mean systolic blood pressure.

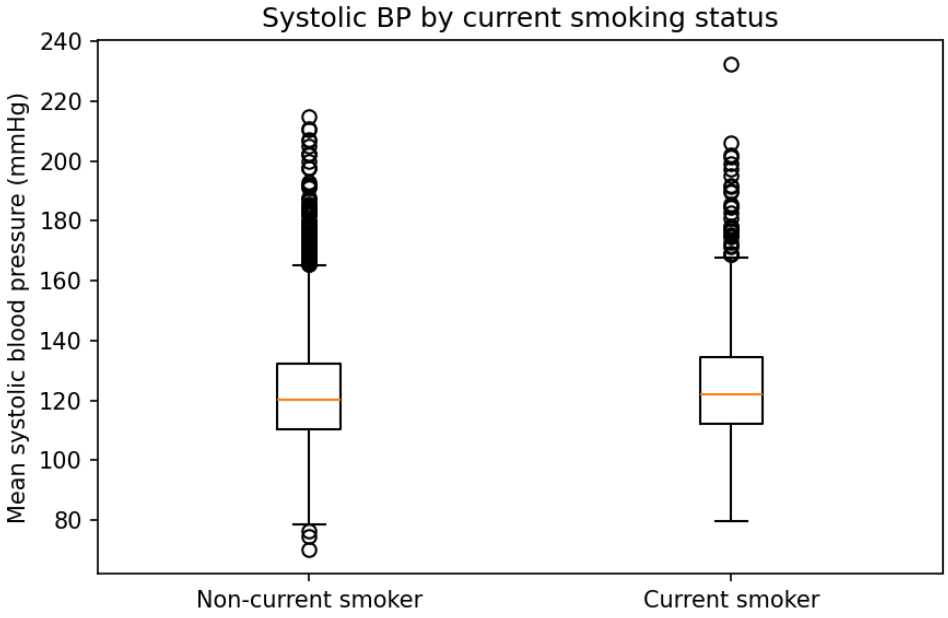


Figure 2. Systolic BP by current smoking status.

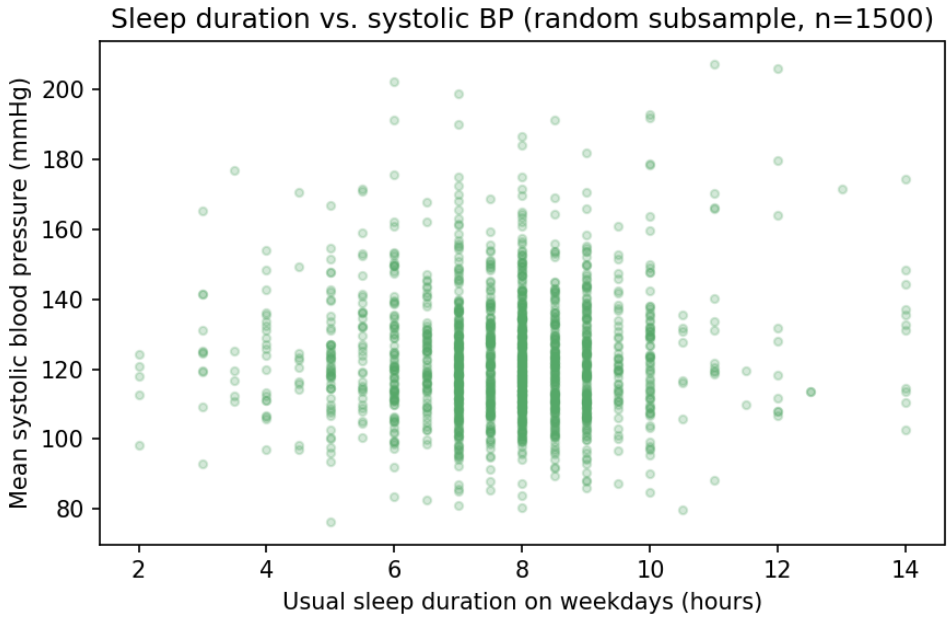


Figure 3. Sleep duration vs. systolic BP (random subsample).

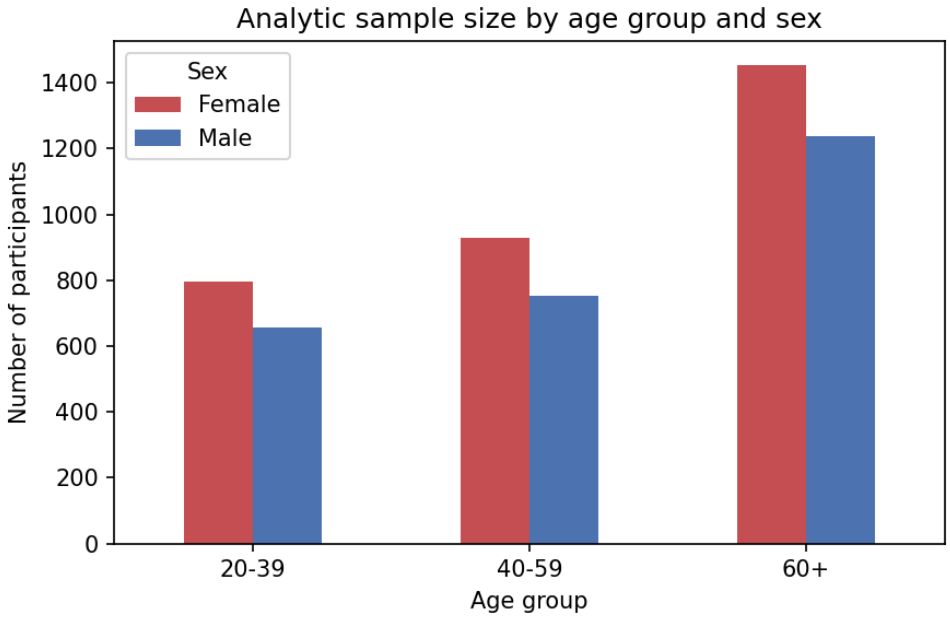


Figure 4. Analytic sample size by age group and sex.

9. Initial Descriptive Comparisons (No Inference Yet)

Only descriptive summaries are compared across groups here; no hypothesis test or p-value is reported in Phase 1 (that is the task of Phase 3).

Systolic BP by smoking group

smoker_group	n	mean	sd	median
Current smoker	877	124.89	19.44	122.0
Non-current smoker	4940	122.71	17.81	120.33

Systolic BP by sex

sex	n	mean	sd	median
Female	3178	121.11	18.85	117.67
Male	2646	125.35	16.84	123.0

Systolic BP by age group

age_group	n	mean	sd	median
20-39	1452	113.52	12.34	112.33
40-59	1681	121.45	16.27	119.67
60+	2691	129.16	19.3	127.0

Track-A (NHANES) Specific Requirements

A1. File Identification and Merge Key

The common merge key across all NHANES files is SEQN. DEMO_L was used as the base file because every participant has a DEMO_L row; the other five files were left-joined on SEQN so no base participant is dropped due to missing data in a sub-component (real exclusions only happen later, in A4, under an explicit rule). No duplicate SEQN values were found in any file.

A2. Identifying the Appropriate Weight and Sampling Design

Because the main response (sbp_mean) and the body-measure variables come from the MEC physical exam (not just the interview), the correct weight is **WTMEC2YR**, not WTINT2YR (which is only valid for interview-only analyses). Using the wrong weight, or ignoring weights entirely, biases population estimates because oversampled respondents represent a different number of population members. The design variables SDMVSTRA (stratum) and SDMVPSU (primary sampling unit) were also identified for correct variance estimation in Phase 3 (15 strata, 2 PSUs observed in this subset).

A3. Raw vs. Weighted Summary Comparison

variable	unweighted_mean	mec_weighted_mean	abs_difference
sbp_mean	123.03	121.27	1.76
bmi	29.73	29.61	0.13
sleep_hours	7.7	7.69	0.0

The unweighted mean treats every respondent equally; the weighted mean weighs each respondent by how many population members they represent ($\sum w_i x_i / \sum w_i$). The gap between the two reflects the effect of the complex (oversampled) design, not a data error.

A4. Defining the Analytic Population

Inclusion criteria:

- Adults aged 20 and older (RIDAGEYR ≥ 20)
- Both interviewed and MEC-examined (RIDSTATR == 2), since blood pressure and BMI require the exam
- At least one valid systolic BP reading

Exclusion criterion: participants who were pregnant at exam time (RIDEXPRG == 1) were excluded, since pregnancy materially alters blood pressure and metabolic physiology and could confound the relationship under study.

Step	N
Merged total (before filters)	11933
Excluded: not fully MEC-examined	3073
Excluded: age under 20	2796
Excluded: pregnant	41
Excluded: no valid BP reading	199
Final analytic sample	**5824**

10. Finalized Statistical Question

Given the analytic sample (N = 5824 adults), good data quality (under 1% missing on most key variables), and an observable descriptive gap between smoking groups, the project's main statistical question is finalized as:

> Is mean systolic blood pressure among U.S. adults aged 20+ (NHANES 2021–2023) associated with usual sleep duration, daily sedentary time, smoking status, and BMI, after controlling for age and sex?

This is a relational / predictive question (addressed with multiple regression in Phase 4), built up through Phase 2 (probability distributions) and Phase 3 (estimation and hypothesis testing). No definitive causal claim is made, since this is a cross-sectional observational study.